

ROYAL PRINCE SINGH

princesinghsikata@gmail.com | +91-8858180597 | Noida, India
github.com/royalprincesingh | linkedin.com/in/royalprincesingh | softprince.dev

PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience building production-ready MERN applications and AI-powered computer vision systems. Proficient in JavaScript, React, Node.js, Python, and MongoDB. Demonstrated ability to design secure authentication systems, optimize backend performance, and implement real-time monitoring solutions. Currently pursuing B.Tech in Information Technology (Expected June 2026). Passionate about writing clean, maintainable code and solving complex technical problems.

TECHNICAL SKILLS

Languages: JavaScript (Advanced), Python (Intermediate), Java (Basic), HTML5, CSS3
Frontend: React.js, React Router, Context API, TailwindCSS, Bootstrap, Responsive Design, Framer Motion
Backend: Node.js, Express.js, REST APIs, JWT Authentication, bcrypt Password Hashing, Middleware
Databases & Cloud: MongoDB (Aggregation Pipelines, Indexing), MySQL, Firebase (Hosting, Firestore, Authentication, Realtime DB)
AI / CV: OpenCV, Python, Real-time Video Processing, Face Detection, Screen Monitoring, Audio Analysis
DevOps & Tools: Git, GitHub, Firebase CLI, npm, Postman, VS Code, MongoDB Atlas, CI/CD Pipelines

EXPERIENCE

Full Stack Intern — Softpro India | Jun 2025 – Aug 2025

AgriTrade: Full-Stack B2B Agricultural Trading Platform (MERN Stack)

- Architected and deployed complete MERN stack application with comprehensive REST API design
- Implemented JWT-based authentication system with bcrypt password hashing for secure user sessions
- Designed MongoDB database schema with strategic indexing on product listings and order collections
- Built Express middleware for role-based access control supporting Admin, Farmer, and Buyer roles
- Optimized database queries and API performance through query refinement and index optimization
- Deployed application on Firebase Hosting with automated CI/CD pipeline using GitHub Actions
- Conducted extensive QA testing validating transaction integrity and data consistency
- Repository: github.com/royalprincesingh/agritrade | Live Demo: agritrade.web.app

Training / Learning Experience — HCL Tech | Jan 2024 – Mar 2025

Java EE Web Development

- Built enterprise-level Java web modules using JSP, Servlets, and JDBC
- Implemented CRUD operations with form-based user authentication and session management
- Designed and optimized MySQL database schemas for reporting and analytics
- Developed SQL queries supporting business intelligence dashboards
- Hands-on experience deploying applications to Apache Tomcat servers
- Gained practical understanding of enterprise software development practices

PROJECTS

Proctor AI — AI-Powered Remote Exam Proctoring System

Technologies: Python, OpenCV, Machine Learning, Flask, React.js, Node.js, MongoDB, Firebase

Repository / Demo: github.com/royalprincesingh/proctor-ai | Live Demo: proctor-ai-exam.web.app

- Engineered comprehensive remote proctoring system with real-time monitoring capabilities
- Implemented face detection and recognition using OpenCV with facial landmark tracking
- Built eye gaze tracking and blink detection for suspicious activity identification
- Designed screen monitoring system detecting application window changes and tab switching
- Created audio monitoring module analyzing voice presence and detecting multiple speakers
- Developed unified violation scoring pipeline with configurable detection thresholds
- Built Flask REST API backend with endpoints for session management and alert logging
- Created React-based admin dashboard for exam management and violation review
- Implemented MongoDB schema for exam instances, alerts, and monitoring logs
- Deployed application to Firebase Hosting with secure authentication

Deepfake Detection System — Computer Vision & Machine Learning

Technologies: Python, OpenCV, Machine Learning, TensorFlow, Real-time Video Processing

Repository / Demo: github.com/royalprincesingh/deepfake-detection

- Developed real-time video analysis system for detecting synthetic media
- Implemented multi-model ensemble approach for improved detection accuracy
- Built 3D pose tracking and skeletal detection for behavioral analysis
- Designed smart behavioral analytics system with anomaly detection
- Achieved production-ready performance for live video stream processing
- Comprehensive feature extraction pipeline and model evaluation framework

Inventory Pro — Enterprise Inventory Management System

Technologies: React.js, Node.js, Firebase (Firestore, Authentication, Hosting), Barcode APIs

Repository / Demo: github.com/royalprincesingh/inventory-pro | Live Demo: inventory-pro.vercel.app

- Built mobile-responsive enterprise inventory tracking system
- Integrated barcode scanning interface for efficient product entry

- Implemented real-time inventory synchronization across warehouse locations
- Designed complex state management for large product datasets
- Created advanced analytics dashboard with inventory insights
- Built multi-warehouse support for distributed inventory management
- Deployed on Vercel with Firebase Firestore for real-time data synchronization

Prince Tech Blog — High-Performance Developer Blog Platform

Technologies: Astro, React.js, Markdown, TailwindCSS, Firebase Hosting

Repository / Demo: github.com/royalprincesingh/prince-blog-boss | Live Demo: softprince.dev

- Built static-site-generated blog using Astro framework for optimal performance
- Implemented automated SEO optimization with meta-tag generation and XML sitemap
- Created responsive design with dark mode support and content categorization
- Designed Markdown-based content workflow for rapid post creation and version control
- Built advanced search functionality across all published articles
- Deployed on Firebase Hosting with automatic CI/CD pipeline via GitHub Actions
- Configured continuous deployment triggering on Git commits to main branch

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology

Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow | Expected Graduation: June 2026

Intermediate (Class 12) — CBSE

P.N. National Public School | 2022

High School (Class 10) — CBSE

P.N. National Public School | 2020

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) Generative AI Professional | 2025

Microsoft Azure AI Essentials | 2025

KEY TECHNOLOGIES & EXPERTISE

- Full Stack Development (MERN Stack)
- Real-time Video Processing & Computer Vision
- AI/ML Integration & Model Implementation
- RESTful API Design & Development
- Secure Authentication (JWT, bcrypt)
- Database Design & Optimization (MongoDB, MySQL)
- Firebase Cloud Platform (Hosting, Firestore, Auth)
- Responsive UI/UX Design
- CI/CD Pipeline Configuration
- Git Version Control & GitHub Workflows

SOFT SKILLS & LANGUAGES

Soft Skills: Communication, Problem-Solving, Team Collaboration, Technical Documentation, Attention to Detail, Quick Learner, Adaptability to New Technologies

Languages: English (Fluent), Hindi (Native)